



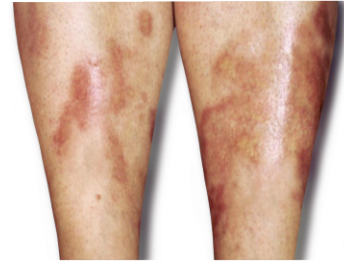
Acanthosis nigricans

- Seen in Insulin resistance and Metabolic syndrome



Necrobiosis lipoidica diabetorum

- Females with T1DM in 3rd decade with atrophic centre containing Telangiectasia



MCQ

Q) A 10-year-old child with T1DM presents to the ER with altered mental status. Diagnosis of DKA is made. O/E thready pulse with acidotic breathing is noted. Labs show a serum potassium of 3.1 mEq/L. What is the most appropriate next step in management?

- a. Normal saline bolus followed by NS drip with KCl
- b. NS infusion for 1 hour along with KCl followed by Insulin 0.1U/kg per hour
- c. Insulin drip with bolus of KCl
- d. Discontinue the use of an Insulin drip due to low Potassium level

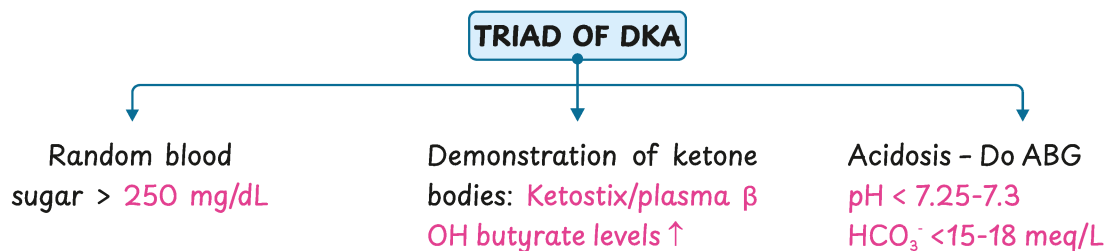
Ans. (a)



DIABETIC KETOACIDOSIS AND HYPERGLYCEMIC HYPEROSMOLAR COMA

DIABETIC KETOACIDOSIS

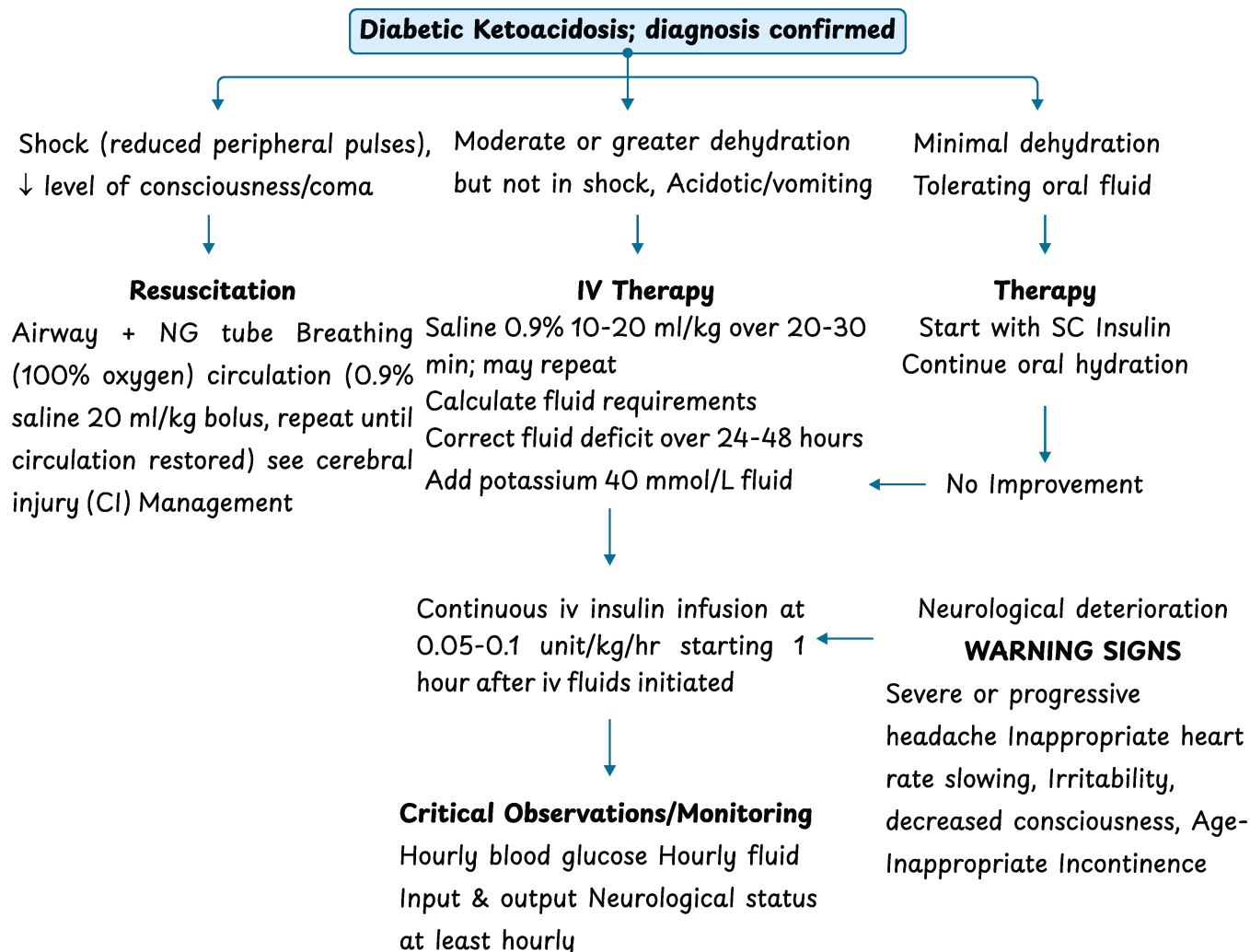
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- **Leading cause** of death in a child with Diabetic Ketoacidosis - **Cerebral edema**
- Highly acidic environment (d/t metabolic acidosis) → damage BBB → Cerebral edema
- Patient presents with stupor, altered sensorium, ↓ level of consciousness, subsequently coma.

ISPAD GUIDELINES

00:09:28



WARNING SIGNS

1. **Neurological deterioration**
2. **Severe, progressive headache**
3. **Decreased consciousness**
4. **Age-inappropriate incontinence**

DIFFERENTIALS FOR AN UNCONSCIOUS DIABETIC BROUGHT TO ER

00:13:49

PARAMETERS	DKA	LACTIC ACIDOSIS	HYPERGLYCEMIC HYPEROSMOLAR COMA	EUGLYCEMIC DKA
pH (7.35-7.45)	7.2	7.2	7.38	7.2
HCO ₃ (22 - 26 meq/L)	15	15	24	15
Ketostix	4+	-ve	-ve	4+
Plasma beta-hydroxy butyrate	↑	Normal	Normal/↑	↑
RBS (mg/dl)	250 - 600	250	600 -1200	200
Plasma osmolality (275 - 295 mOsm/kg)	310	300 Plasma lactate- ↑	330-380	295

LACTIC ACIDOSIS

- Ketone values normal
- Confirmed by blood culture, metagenomic sequencing (if available), check if patient has gone into septic shock, KFT & plasma lactate values.
- Reason:
 - Mis prescription on part of physician, eGFR low, still metformin given
 - Septic shock

HYPERGLYCEMIC HYPEROSMOLAR COMA

- ↑ blood sugar levels in blood → cross BBB (create osmotic gradient) → Water enters brain → cerebral edema → coma

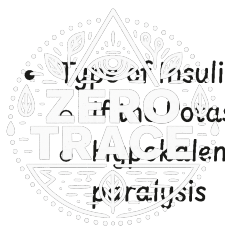
EUGLYCEMIC DKA

- Patients are concomitantly on SGLT2 inhibitor (canagliflozin or Empagliflozin) for diabetes → cause glucosuria (loss of sugar in urine) → Sugar value is on lower side of threshold

MUST KNOW FACTS

00:25:59

- **Fruity odor** of breath is due to **ketone body (acetone)**.
- Limitation of Ketostix: measure Acetoacetate (can be falsely normal)
 - As acetoacetate degrades into Acetone
- **IOC for DKA:** Plasma β OH butyrate ↑
- **Role of Soda bicarbonate in DKA management - No role**



- Type of insulin used - **Regular insulin IV** → Send potassium into cells
 - if the potassium value is < 3.3 meq/L then hold the insulin drip
 - Hypokalemia predisposes to arrhythmias, muscle cramps, respiratory muscle weakness, diaphragmatic paralysis

PATHOPHYSIOLOGY OF DKA

00:31:13

Deficiency of insulin



Insulinopenia



Subcutaneous fat oxidation → **weight loss**



Generation of free fatty acids



Generation of Ketones



Decline in pH (Acidic pH) → Stimulates respiratory center → **Kussmaul breathing**



Damage to BBB



Cerebral edema → ↓ **level of consciousness**

TRIGGERS FOR DKA

- Infection
- Myocardial Infarction (d/t Type 2 DM)
- Insulin pump malfunction, Poor compliance with insulin injection
- Cocaine use
- Surgery

MORTALITY RATE

- > 40 years: > 20%
- < 40 years: ~ 5%

CLINICAL FEATURES

00:37:33

- Type 1 DM child with insulin stopped/undiagnosed child with pneumonia
 - Nausea, Vomiting: as Ketonemia stimulates CTZ
 - Epigastric pain/Acute abdomen
 - Stupor/Coma: d/t Fluid shift across the brain
 - Polyuria: d/t Osmotic diuresis
 - Dehydration – d/t vomiting & polyuria
 - Encephalopathy
- Examination findings
 - Fruity odor in the breath
 - Pulse - **Thready pulse**/Fast pulse
 - ↓ BP
 - Kussmaul breathing



TESTS	DESCRIPTION
Urine ketostix	4+
Plasma β -hydroxybutyrate levels	$\uparrow$
Random blood sugar	300 mg/dL
Serum Potassium	$\downarrow$
Serum Sodium	$\downarrow$ - Solvent drag $\rightarrow$ Hypertonic hyponatremia $\uparrow$ sugar $\rightarrow$ draw water $\rightarrow$ Dilution
Kidney function test	$\uparrow$
ABG	- pH $\downarrow$ - pCO₂ $\downarrow$ - HCO₃⁻ $\downarrow$ Partially compensated metabolic acidosis
Anion gap	$\uparrow$ High anion gap metabolic acidosis
Serum amylase and lipase	Normal/ $\uparrow$
Total leukocyte count	$\uparrow$
Blood culture	Diagnose sepsis

Important Information

- Hypervolemic Hyponatremia - TBW $\uparrow$, TBS $\uparrow$
- $\uparrow$ aldosterone can be seen in CHF, AKI, CKD, Cirrhosis

- For every 100mg/dl $\uparrow$ in sugar value, there will be 1.6 meq/L $\downarrow$ in Na.

CONFIRM DKA BASED ON	- Blood sugar ↑ - Ketonemia/Ketonuria - HCO₃⁻ < 15-18 meq/L				
ADMIT TO ICU IF	- pH < 7.0 - Extensive Labored breathing - Impaired level of arousal				
ASSESS					
Na+	Potassium	Mg	PO ₄	Cl-	HCO ₃ ⁻



↓	Cause • Vomiting • Insulin drip	↓ On treatment	↓ On t/t	↑ Shift from 0.9% NS → 0.45% NS	↓
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- NS excess → Hyperchloremic acidosis
- Chloride excess → ↑ negative charges in blood
- Kidney tries to maintain Electroneutrality → Send HCO_3^- inside the cells → ↓ HCO_3^- (Acidosis worsens)

GRADING OF DKA

00:57:34

STAGE	pH	β -OH Butyrate
Mild	7.25 - 7.30	3 - 4 mmol/L
Moderate	7.24-7.00	4 - 8 mmol/L
Severe	< 7	> 8 mmol/L

- Lower is the value of bicarbonate → Severity of DKA

ONE-LINERS ON DKA

00:59:24

1. Hyperglycemia improves at the rate of 50-100 mg/dL per hour
2. When plasma glucose reaches 200-250 mg/dl: Give 5% dextrose + 0.45% saline
3. Start insulin as soon as the patient is eating give Subcutaneous insulin
4. Leading cause of death in a child with DKA → **Cerebral edema**
5. Causes of death in adults with DKA - **ARDS, Renal failure, Bowel infarction**
6. First-line management in DKA - NS 0.9% + KCL 40 meq/L or RL
7. TOC in DKA: **Regular Insulin infusion**
8. SC Insulin can be given in mild DKA - Yes
9. Role of bolus insulin /soda bicarbonate - Not recommended

HYPERGLYCEMIC HYPEROSMOLAR COMA

01:03:48

CLINICAL FEATURES	• Occult diabetes in middle-aged patients/T2DM • When sugar is high → polydipsia & Polyuria → Dehydration • Insidious onset pre-renal AKI causing azotemia ◦ Dehydration → blood urea nitrogen & creatinine value ↑ → Prerenal AKI → Azotemia ◦ ↑ Dehydration → blood supply to Kidney decreases • ↓ level of consciousness ◦ ↓ cerebral blood flow ◦ Uremia (Azotemia + clinical features)
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Relative insulin deficiency causing hyperglycemia that triggers osmotic diuresis

TRIGGERS

- MI
- Stroke
- Pneumonia (Covid-19)
- Surgery

WORK UP

- **RBS: 600 to 1200 mg/dL**
- NCCT head: normal
- ABG: **Normal pH, PCO₂, and HCO₃⁻**
- Plasma beta-hydroxybutyrate: **Normal/ ↑**
- **Plasma osmolality:** 330 to 380 mOsm

TREATMENT

- **Normal saline 0.9%** (10 L over 48 hours)
- Sodium > 150 meq/L → Give 0.45% Normal saline
- Insulin drip is continued till the patient begins to eat



- Q. A child is brought to the ER in an unconscious state after severe vomiting and abdominal pain episodes. HR = 100/ min and BP = 100/60 mmHg. The work-up showed pH = 7.2, blood glucose = 300 mg/dl, and ketone = 3+. What should be the best management?
- a. An **Infusion** of 0.05 to 0.1 U/kg dose in insulin after 1 hour of IVF
 - b. 0.1 unit/kg insulin stat followed by infusion of 0.5 to 0.1 U/kg Dose insulin and IVF
 - c. 0.1 U/kg insulin subcutaneous stat followed by IVF
 - d. IVF followed by subcutaneous 0.1 U/kg insulin

Ans (a)

- Q. Which of the following is not a diagnostic criterion for Diabetic ketoacidosis? (INICET)
- a. pH < 7.3
 - b. Serum Potassium < 3.5 mEq/L
 - c. Serum bicarbonate < 15 mEq/L
 - d. Serum glucose > 250 mg/L

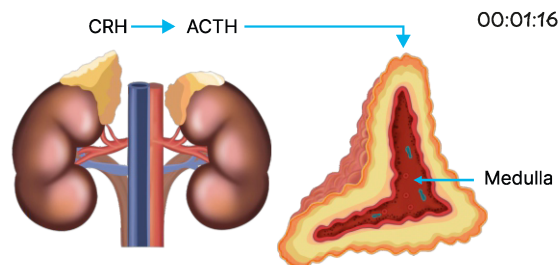
Ans (b)



DISEASES OF THE ADRENAL GLAND

PHYSIOLOGY OF ADRENAL GLANDS

- Adrenal glands are asymmetrical
 - Right - pyramidal
 - Left - semilunar
- Hormone from the hypothalamus that regulates ACTH → CRH (corticotropin-releasing hormone)



LAYERS OF ADRENAL GLAND

ADRENAL CORTEX

LAYERS	DESCRIPTION
ZONA GLOMERULOSA	- Responsible for Producing Aldosterone - Aldosterone - Acts on ENaC (Epithelial Na²⁺ Channels) located in Collecting duct >> DCT - Main function of aldosterone - Postural regulation of blood pressure - Mechanism - Salt and water retention → Acts on ENaC (Epithelial Na²⁺ Channels) located in Collecting duct >> DCT → Corresponding loss of potassium as well as hydrogen. - Other function - Maintain pH, potassium in body. 1° control of Aldosterone is by RAAS system (JG cells)
ZONA FASCICULATA	- Responsible for the release of cortisol - Cortisol → stress hormone → regulates blood sugar levels
ZONA RETICULARIS	- Releases sex steroids like DHEAS (Dihydro Epiandrosterone Sulfate) - Responsible for secondary sexual characters

ADRENAL MEDULLA

- Synthesize & release catecholamines - Epinephrine, Norepinephrine, Dopamine
- Major circulating catecholamine - **Epinephrine** (60%)
- Catecholamines are responsible for fight/flight phenomenon
- Major circulating catecholamine in pheochromocytoma - Norepinephrine
- Pheochromocytoma outside adrenal medulla → Paraganglioma
- 1° action of ACTH is on zona fasciculata of adrenal gland
- Vanillylmandelic Acid (VMA) - end product of the metabolism of catecholamines
- 24-hour urinary VMA levels → accurate for the diagnosis of Neuroblastoma > Pheochromocytoma
- Test useful for diagnosis of pheochromocytoma - urinary metanephrine levels.
- AA responsible for synthesis of catecholamine → Tyrosine (also for T₃, T₄)



HYPERALDOSTERONISM

00:12:17

PRIMARY HYPERALDOSTERONISM	SECONDARY HYPERALDOSTERONISM
Bilateral Adrenal Hyperplasia - m/c cause (90%) CONN'S Syndrome: Autonomous Tumor in 1st layer of adrenal gland. - Produce huge amount of aldosterone → HTN, Hyperkalemic alkalosis.	Massive ascites Redistribution of fluid to peritoneal cavity ↓ Circulating fluid volume ↓ ↓ GFR ↓ ↓ Stimulation of RAAS ↓ Aldosterone ↑ Others-Pregnancy-induced HTN, Bartter syndrome, Gitelman syndrome

Important Information

- Causes of Ascites – Cirrhosis, CHF, Pericarditis, Nephrotic syndrome

CLINICAL FEATURES OF PRIMARY HYPERALDOSTERONISM

00:18:14

- Hypertension with hypokalemic alkalosis**

Increase in aldosterone



Increase in salt & water absorption



Expansion of plasma volume



Stretching of right atria



Atrial natriuretic factor release

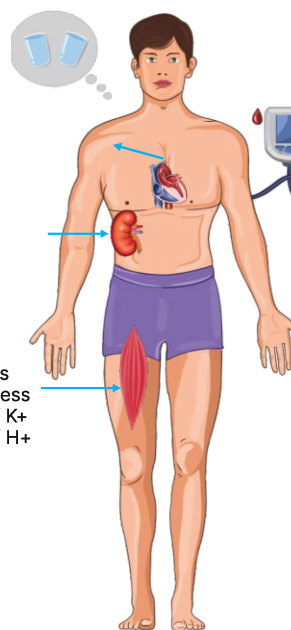


Salt and water loss



No Pedal Edema

- Polyuria
- Polydipsia



- Muscle Cramps
- Muscle Weakness
- Urinary loss of K⁺
- Urinary loss of H⁺

- Headache
- HTN Resistant
- HTN Persisting despite ≥ 3 anti HTN medication

WORK UP

00:27:02

- Sodium levels: ↑ / normal
- Potassium levels -↓
- Screening test : Aldosterone: Renin Ratio (ARR) elevated (> 750 pmol/L)**
 - False positive d/t β-blockers



- NCCT Abdomen – Bilateral adrenal hyperplasia
- **Adrenal vein sampling** → used for localization of tumor (pre-operative)
- **LC** – Same infusion test, Salt loading test
 - Interpretation for salt infusion test – Not be able to suppress aldosterone production

Important Information

- Na^+ and K^+ have inverse relation in the presence of aldosterone
- Calcium and PO_4 have an inverse relation in the presence of Parathyroid hormone
- Plasma osmolality and urine osmolality have an inverse relation in the presence of ADH
- Low renin hypertension is seen in Conn's syndrome (tumor → ↑ aldosterone → negative feedback to RAAS), Liddle syndrome

TREATMENT

00:33:03

BILATERAL ADRENAL HYPERPLASIA	CONN'S SYNDROME
• T/t - Eplerenone, Spironolactone ◦ Risk of gynecomastia in male (more with spironolactone)	• Medical management: Spironolactone (for 6 weeks) • After preanesthetic checkup & fitness → U/L Adrenalectomy

CONNSHING SYNDROME

00:35:08

- Tumor in both 1st and 2nd layer of adrenal gland.
- Causes ↑ Aldosterone (Hypertension with hyperkalemia) & ↑ Cortisol (hyperglycemia).

ADRENAL INCIDENTALOMA

00:36:05

- Chance detection on related imaging
- First, suspect it to be a malignancy → Rule it out by doing PET Scan or whole-body PET CT → monitor for biochemical excess of a hormone → Prove it to be non-functioning (No hormone overproduction)

LIDDLE SYNDROME

00:37:54

INHERITANCE	• Autosomal dominant
DEFECT	• ENac exhibits a gain of function
PATHOPHYSIOLOGY	<pre> graph TD A[GAIN OF FUNCTION] --> B[Gain of salt and water] A --> C[Exaggerated loss of potassium and hydrogen ions] B --> D[Expansion of plasma volume] D --> E[Hypertension] C --> F[Hypokalemic metabolic alkalosis] E --> G[Negative feedback to the RAAS system] F --> G G --> H[↓ Renin, Aldosterone] </pre>



DIAGNOSTIC TEST

- Aldosterone Renin Ratio: Normal/↓
- NCCT abdomen - Normal adrenal gland

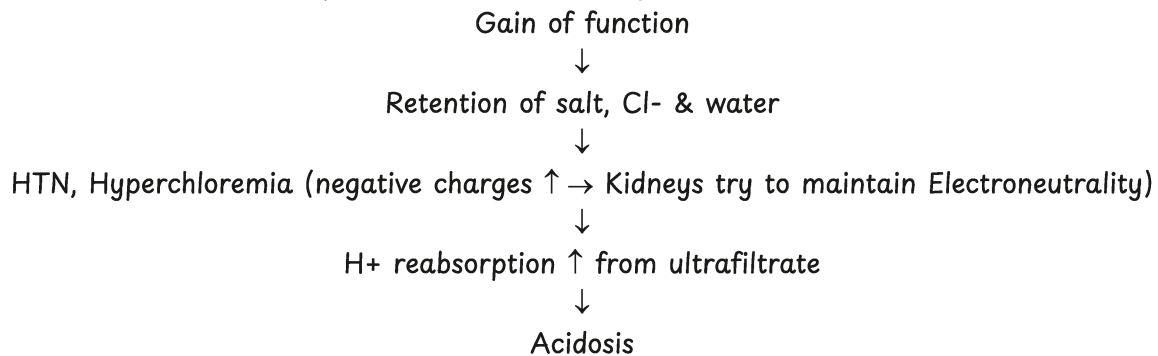
TREATMENT

- DOC - **Amiloride**

GORDON SYNDROME

00:43:06

- Defect: Sodium chloride cotransporter in DCT exhibits a gain of function



HTN WITH HYPOKALEMIC ALKALOSIS

Conn's syndrome, Liddle syndrome & Cushing's syndrome

HTN WITH HYPERCHLOREMIC ACIDOSIS

Gordon Syndrome

PRIMARY ADDISON'S DISEASE

00:46:50

- Damage to both adrenal glands
- All 3 layers of adrenal cortex are severely affected

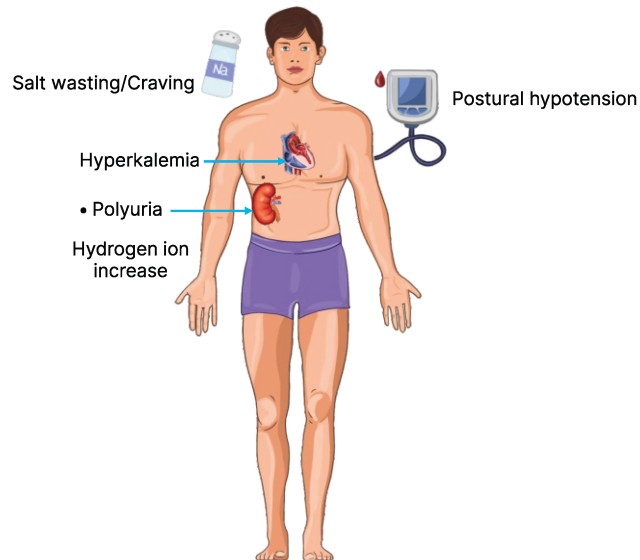
ETIOLOGY

00:47:15

- **Isolated autoimmune adrenalitis - M/c cause**
- Autoimmune polyglandular syndrome type 1:
- **Manifestation**
Mnemonic - CHAI
 - **C**andidiasis (Mucocutaneous) - d/t involvement of thymus gland
 - **H**ypoparathyroidism
 - **A**ddison disease - **AIRE gene on chromosome 21**, Autosomal recessive
 - **I**nfancy manifestations
- Infectious causes
 - TB
 - **HIV**
 - Histoplasmosis
 - Sarcoidosis - non caseating granulomas
 - Adrenoleukodystrophy
- **Addisonian crisis** - Aka **Waterhouse Friedson syndrome**
 - Seen after a fulminant infection by N. Meningitis leading to sepsis, DIC
 - Internal bleeding in adrenal gland → ↓ Aldosterone & BP
 - Main T/t - **Intravenous hydrocortisone**

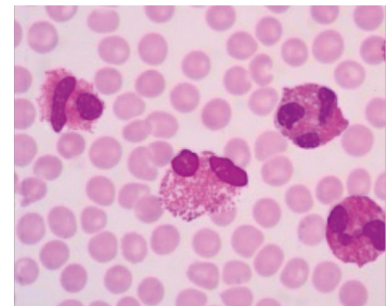
DECREASED ALDOSTERONE

- **Postural hypotension**
 - Initially check BP in supine position
 - On changing the position from supine to standing, **SBP falls by ≥ 20 mmHg & DBP falls by ≥ 10 mmHg in 3 min.**
- **Hyperkalemia** – affect cardiac conduction $\rightarrow$ cardiac arrest



DECREASED CORTISOL (Isolated cortisol deficiency)

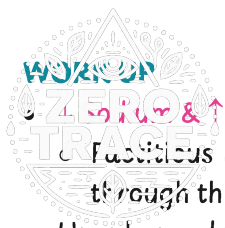
- **Hypoglycemic attacks**
- Emotional lability
- Dizziness on sudden change of position
- Craving for salty foods
- Loss of weight
- Loss of inhibitory control over vasopressin $\rightarrow \uparrow$ ADH $\rightarrow$ water gain $\rightarrow$ **Hyponatremia**
- Eosinophilia



Eosinophilia

LOSS OF SEX STEROIDS Low DHEAS

- Fatigue
- Loss of Libido
- Erectile dysfunction
- Loss of pubic hair, and axillary hair in females
- **Hyperpigmentation** - d/t $\uparrow$ ACTH (partial MSH activity)
 - Knuckles, Interphalangeal joint
 - **Palm/sole creases**
 - Axilla, Groin
 - Lips
 - Oral mucosa



01:03:26

↓ Potassium levels

- Pseudohyperkalemia - Avoid sampling taking using a thin bore needle as RBCs can burst travelling through thin bore needle → false elevation of potassium.
- Unenhanced CT/NCCT abdomen - Reveal damage to both the adrenal glands
 - **IOC - ACTH stimulation test/ Cosyntropin test** → Fail to stimulate gland

TREATMENT

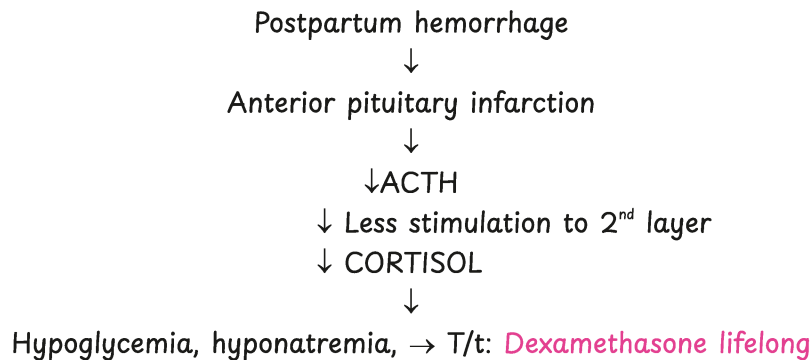
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- Drug of choice: **Hydrocortisone (orally)** → Lifeline
 - Have both mineralocorticoid & glucocorticoid activity
- Additional drugs:
 - Low BP - Fludrocortisone (predominantly Mineralocorticoid)
 - Low sugar - Dexamethasone (Glucocorticoid)

SECONDARY ADDISON'S DISEASE

SHEEHAN SYNDROME

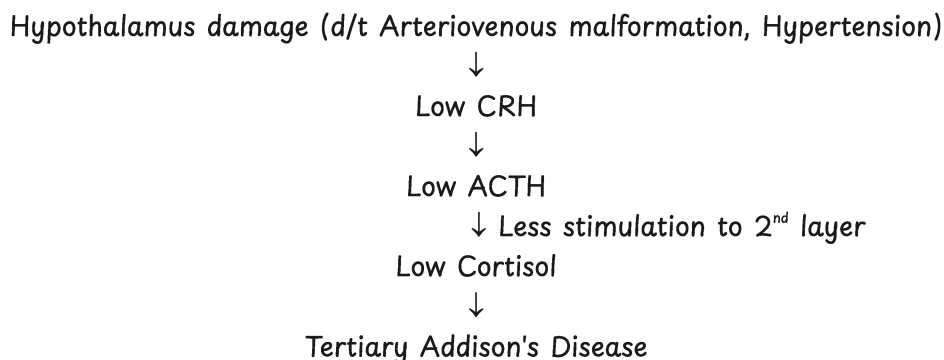
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- ↓ ACTH → **Ivory white skin/ Pale alabaster skin** (Hypopigmentation)

TERTIARY ADDISON'S DISEASE

01:10:04



TREATMENT

- **Dexamethasone (DOC)**

FACTS ABOUT PRIMARY, SECONDARY & TERTIARY ADDISON'S DISEASE

	CAUSE	ACTH	HYPERPIGMENTATION (Palm/ Sole creases)	DOC
1°	Autoimmune	↑		Hydrocortisone
2°	- Sheehan syndrome - Non-obstetric damage to the pituitary → Simmond disease	↓	-	Dexamethasone
3°	Hypothalamus damage	↓	-	Dexamethasone
CRISIS	Waterhouse-Friedson syndrome Sudden stoppage of Steroids	Normal	-	I.V. Hydrocortisone

CONN'S VS ADDISON'S DISEASE

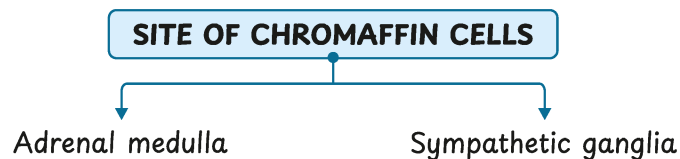
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	CONN'S Syndrome (↑ Aldosterone)	ADDISON'S DISEASE (↓ Aldosterone)
Cause	Adrenal Adenoma	Autoimmune destruction of adrenals
BP	Hypertension	Postural hypotension
K ⁺	↓	↑
pH	Alkalosis	Normal
Special finding	No Pedal edema	Hyperpigmentation (on frictional areas)
IOC	Saline infusion test	ACTH stimulation test or Cosyntropin test
Treatment	Spironolactone	Hydrocortisone

PHEOCHROMOCYTOMA

01:17:41

- Aka Chromaffinoma



- Catecholamines (normal person) → Norepinephrine (80%) > Epinephrine (20%) > Dopamine
- Pheochromocytoma → Majorly produce norepinephrine
- Pheochromocytoma in association with MEN Type 2A/Sipple syndrome → Majorly Produces epinephrine
- Paraganglioma – Extra-adrenal catecholamine-producing tumor
 - Common site – Skull base
- Benign vs Malignant tumor



10% cases can be malignant

Biopsy/FNAC → Contraindicated → as it may result in damage to regular layers of the gland

Differentiated by PET/CT: Ga 68 Dotatate - Imaging modality of choice

LOCATION OF TUMOR

85%-90% of cases - Inside medulla

10-15% of cases - Outside medulla

Location of sympathetic ganglia - Organ of Zuckerkandl

10% cases have Familial basis and association with MEN syndrome type 2A/Sipple syndrome

Any 2 of the following 3 is sufficient for diagnosis of MEN

A 20-year-old patient came with hypertension which was difficult to manage. Work up - 24-hour urinary catecholamines & 24-hour urinary metanephrine were elevated.

→ He was diagnosed with Pheochromocytoma. Surgery was done.

→ Intraoperative and postoperative periods were uneventful.

→ The patient was discharged

The same patient at 40 years old came with recurrent kidney stones.

→ Blood work showed Serum calcium, Serum PTH - ↑

→ Sestamibi Tc99 SPECT/CT showed Parathyroid adenoma

→ T/t for big & multiple kidney stone - PCNL, if < 2cm do lithotripsy.

→ To prevent recurrence - Superficial parathyroidectomy done

The same patient at 60 years old came with stony hard lump in neck & was diagnosed with Medullary carcinoma thyroid (calcitonin)

Cause - Abnormal gene (ret gene) on chromosome 10

10% cases are bilateral.

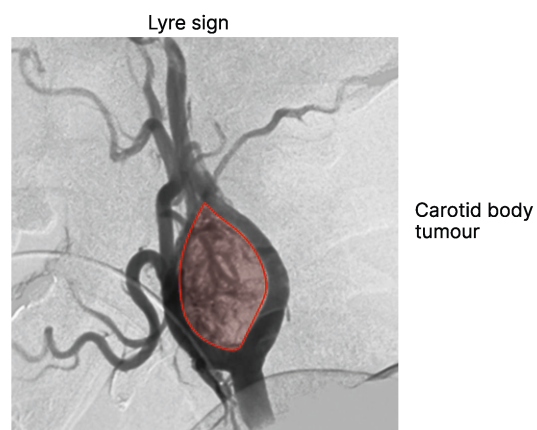
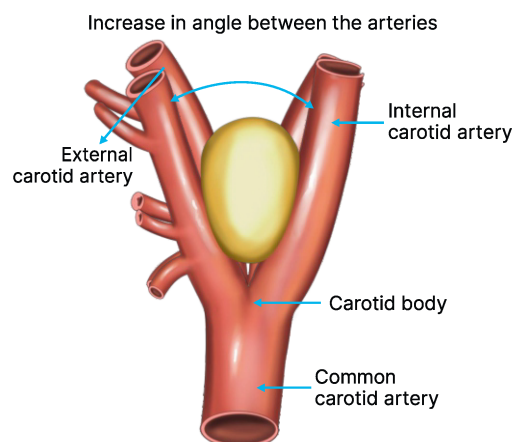
PARAGANGLIOMA

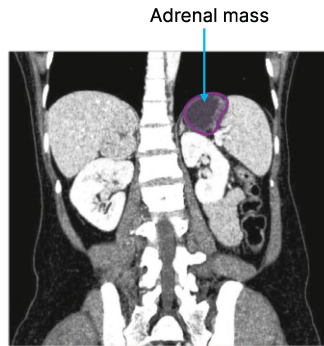
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Origin - Carotid body

Carotid body - peripheral chemosensor innervated by Vagus nerve: measure O_2 & CO_2 levels in common carotid artery

As tumor grows → angle between ECA & ICA increases → Lyre Sign.





- **Nests of Zellballen** - Histopathology finding in Paraganglioma

CLINICAL FEATURES OF GREAT MASQUERADER

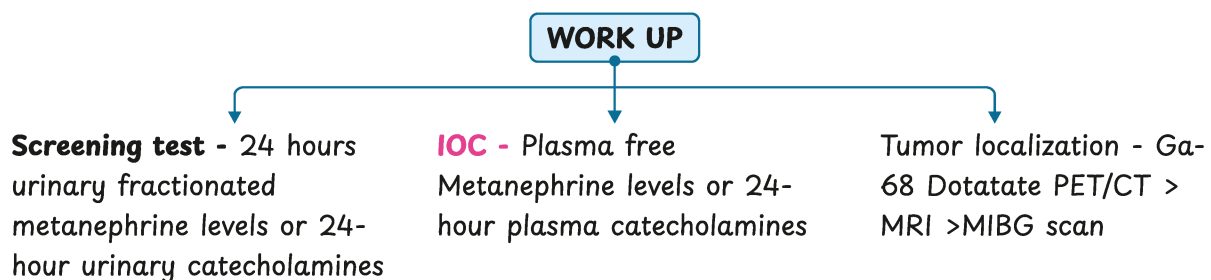
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- Misdiagnosis - Anxiety disorder
- Palpitations - d/t β_1 activation
- **Headache** - d/t α_1 activation
 - Vasoconstriction - **Paroxysmal hypertension, Episodic hypertension**
 - Can be triggered by position change, exercise, urination
- Tremors and sweating in palms & feet (Diaphoresis) - d/t β_2 stimulation in muscles
- Blood sugar
 - Fasting blood glucose - 100-125 mg/dL
 - 2-hour plasma glucose - 140-199 mg/dL
- Weight loss
- Exaggerated vasoconstricted over longer duration of time → Volume contracted state achieved → Postural hypotension
 - Lab parameters - **relative polycythemia, hypercalcemia**

Important Information

- Norepinephrine - Shortest half life

01:42:53



TREATMENT

01:45:30

- TOC - **Surgery in Benign** Pheochromocytoma
 - Partial /Total adrenalectomy
- **Surgery is contraindicated** in **malignant pheochromocytoma**
 - Radiation (radioactive iodine), chemotherapy



PREOPERATIVE HTN MANAGEMENT

INTRA-OPERATIVE HTN MANAGEMENT DURING PHEOCHROMOCYTOMA SURGERY

- For HTN - **Oral Phenoxybenzamine (at 6 am)**
- For tremors & palpitations - **Propranolol (Post breakfast)**
- If propranolol is taken 1st → Unopposed α action → further spike in hypertension
- $\alpha + \beta$ → recommended
- $\beta + \alpha$ → contraindicated

- IV Sodium nitroprusside
- Phentolamine

CARCINOID TUMOR VS PHEOCHROMOCYTOMA

01:53:46

	CARCINOID TUMOR	PHEOCHROMOCYTOMA
Site	• Bronchus > ileum, rectum, appendix	• Adrenal medulla, sympathetic chain, carotid body
Symptoms	Symptoms are due to excess of 5HT i.e., Serotonin & Histamine • Flushing episodes over face, neck or upper trunk • Secretory diarrhea • Asthma-like illness • Pruritus • Gastritis • Pellagra (Niacin deficiency → Tryptophan is utilized for 5HT synthesis) • TIPS (Tricuspid Insufficiency & Pulmonary Stenosis) Flushing  Pruritus 	• Palpitation • Headache • Diaphoresis } Due to excess catecholamines
Testing	• 24-hour urinary 5 HIAA (Hydroxy Indole Acetic Acid) levels	• 24-hour urinary fractionated metanephrine levels
Imaging modality	• PET/CT - Gallium 68 Dotatate • Octreo scan	• PET/CT - Gallium 68 Dotatate • MIBG scan
Treatment	• Octreotide • Telotristat	• Phenoxybenzamine • Sodium nitroprusside



Q. A 42-year-old female presents with a history of fatigue, weight loss, and hyperpigmentation on knuckles and palm creases. Laboratory investigations reveal low sodium and high potassium levels. An unenhanced CT of the abdomen shows destruction of bilateral adrenal glands. Which investigation is the most appropriate to confirm the diagnosis of this patient?

- a. Aldosterone-renin ratio
- b. Cosyntropin test
- c. Saline loading test
- d. 24-hour urinary control values

Ans (b)

Q. All of the following are features of Conn's syndrome except?

- a. Metabolic alkalosis
- b. Metabolic acidosis
- c. Hypertension
- d. Muscle cramps

Ans (b)

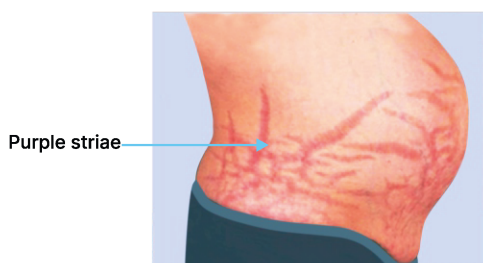


12. CUSHING SYNDROME

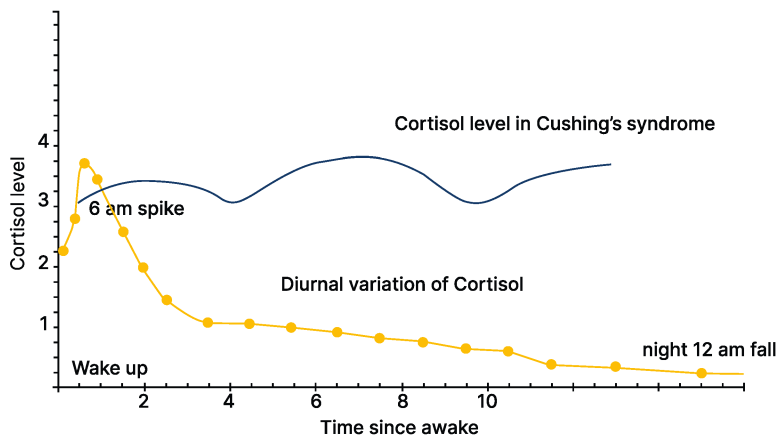


BASELINE PHYSIOLOGY

00:01:19



Purple striae



DIURNAL VARIATION OF CORTISOL

00:01:20

- Cortisol → released in pulsatile fashion
- Spike : 6 am -12pm
- 75% of daily cortisol is produced in morning hours
- Cortisol lowest at 12 AM (midnight)
- **Earliest manifestation** of Cushing's syndrome : Loss of diurnal variation of Cortisol production
- M/c cause of Cushing's syndrome is **iatrogenic steroids (exogenous cause)**
- M/c cause of ACTH-dependent Cushing's syndrome - **Pituitary adenoma/aka Cushing disease (Endogenous cause)**

CAUSES

00:06:41

CAUSES	DESCRIPTION
• Iatrogenic steroids	Cortisol ↑, ACTH ↓ by feedback mechanism
• ACTH-excess-producing pituitary adenoma ◦ Endogenous cause ◦ Cushing disease (F: M =4:1) → More common in ACTH dependant	ACTH ↑, Cortisol ↑
• Ectopic production of ACTH	Carcinoid tumor (Bronchus, Pancreas), Oat cell carcinoma lung, Medullary thyroid cancer, Pheochromocytoma
• Adrenal Adenoma	• Cortisol ↑ , ACTH ↓ • ACTH-independent Cushing syndrome • Autonomous